

Arpan Pathak

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SUMMARY

Systems Software Engineer with **8 years** architecting high-performance, low-latency systems at **Amazon**, **Microsoft**, and **Oracle** – database internals, kernel-level networking, distributed systems, and real-time ML inference. Deep expertise in **Rust**, **C++**, **Go**, and **C**: lock-free concurrency, memory safety, eBPF/Cilium, and GPU-accelerated computing. Shipped measurable wins: **~40% lower P99 latency**, **2x throughput**, **45,000 ops/s** throughput gains, **5,000 QPS** inference serving, and **\$5.4M/month** revenue impact.

PERSONAL PROJECTS

Driving-CivicSense (Edge AI Vision) | github.com/arpanpathak/driving-civicsense-vision-model | 2026

- On-device AI perception in **Rust** (intersection discipline, lane awareness) with **< 50ms inference** and zero cloud dependency on Raspberry Pi Zero / Pi 5; fine-tunes **YOLOv8/v11** + custom **CNN** with pruning & quantization for tight memory/power budgets.

cuda-oxide (Rust-to-CUDA Compiler) | github.com/arpanpathak/cuda-oxide

- Experimental **Rust-to-CUDA** compiler: write SIMT GPU kernels in idiomatic Rust and compile straight to **PTX** – no DSLs, no bindings.

Companion Book: Seeing Machines: Deep Learning & Computer Vision from Python to Bare Metal | arpanpathak.github.io/seeing-machines-book

Physics & Mathematics of Game Development in Rust (Book) | arpanpathak.github.io/bevy-physics-book | 2026

CUDA Kernels: GPU & Parallel Programming from First Principles (Book) | arpanpathak.github.io/gpu-parallel-book | github.com/arpanpathak/gpu-parallel-book | 2026

EXPERIENCE

Software Developer II, Microsoft, Redmond, WA

Jun 2025 - Present

- C++/C# → Rust Migration (Memory Safety)**: Spearheaded migration of the Information Protection service from **unsafe C++** and **C#** to **Rust**, moving memory safety to compile time – eliminated use-after-free, buffer-overflow, and null-safety bugs in the migrated path. Delivered **~40% lower P99 latency**, **2x throughput**, **3x lower memory** per request, and **6-figure annual cost savings**.
- Zero-Trust Kernel Networking (eBPF/Cilium)**: Architected zero-trust sovereign cloud networking with **eBPF + Cilium** network policies enforcing **data residency** and **SecNumCloud** compliance; authored **Kubernetes operators (Kubebuilder)** to reconcile policy state across AKS clusters in France and Germany.
- AI-Powered Policy Engine**: Built an AI engine that auto-generates Cilium policies and scans for misconfigurations – manual review cut from **4 hours to near-real-time**, zero policy drift across sovereign regions.
- CI/CD & Observability**: Containerized microservices with blue-green deployments, **Prometheus/Azure Monitor**, and automated validation for reliable multi-region releases.

TechStack: Rust, C++, eBPF, Cilium, Kubernetes (AKS), Kubebuilder, Prometheus, Linux

Senior Software Engineer, Oracle Cloud Infrastructure, Seattle, WA

Oct 2024 - Mar 2025

- High-Performance Database Internals**: Architected low-latency internals for a proprietary database engine, optimizing **lock-free data structures** to cut contention in high-concurrency read-write paths – **45,000 ops/s** transaction throughput gain.
- Infrastructure-as-Code SDK**: Engineered a **Terraform provider** (developer-facing SDK) for Oracle Autonomous Database, replacing legacy control-plane tooling with **\$1.2M/month** projected savings.
- Security & Key Management**: Integrated OCI Security Vault across control-plane services for secret management and automated key rotation.

TechStack: Rust, C++, Go, Terraform, PostgreSQL internals, OCI

Software Development Engineer II, Amazon, Seattle, WA & Hyderabad, India

Mar 2021 - Oct 2024

- Real-Time ML Inference Engine**: Architected **XGBoost + Deep Learning ensemble** ranking for ads & deals across FreeVee, Twitch, Prime Video, and Amazon Retail – millions of events at **100ms inference latency**, driving **5% CTR lift**, **30%** more deal impressions, and **\$5.4M/month revenue growth**.
- BERT Inference Service**: Delivered **BERT-based** NLP inference handling **5,000 QPS** at peak, cutting support-ticket triage SLA from **7 days to 2 hours**.

- **Distributed Systems & SDK Design:** Designed uniqueness-constraint indexing and an **SDK** for a multi-tenant NoSQL store using **two-phase commit**, achieving **serializable isolation** at **3K writes/second** – a developer-facing library used by multiple publisher teams.
 - **Event-Driven Ads (CDC):** Delivered programmatic guaranteed ad deals via real-time **change data capture (CDC)** with **< 50ms latency** for auction and programmatic buying platforms.
 - **Big Data Platform:** Built **1-petabyte** data lake & warehouse with **Apache Spark, AWS Glue**, Kotlin Spark API, and a **DSL execution engine** for OLTP and BI use cases.
- TechStack:** *Kotlin, Java, Python, PyTorch, XGBoost, Spark, Kafka, DynamoDB, Redshift, AWS*

Software Development Engineer, Razorpay, Bengaluru, India

Aug 2020 - Feb 2021

- Built UPI Payment Gateway Aggregator in **Go** + Protobuf processing **1M+ hourly transactions** with **99.99% uptime** at **150ms P99** under **50,000 TPM** peak load.
- TechStack:** *Go, Redis, PostgreSQL, Protobuf, Kafka*

Software Engineer, Mindfire Solutions, Bhubaneswar, India

Aug 2018 - Feb 2020

- Delivered full-stack workflow tools and an **AR-based product search microservice** serving **10,000+ daily queries**; cut dashboard latency from **5 min to 1 min**.
- TechStack:** *Java, Spring Boot, Angular, MySQL, MongoDB*

SKILLS

Languages: Rust, C++, Go, C, Python, Java, Kotlin, TypeScript, Shell Script

Systems Programming & Performance: Lock-Free Data Structures, Concurrency, Memory Safety, Memory Profiling, Cache Optimization, Performance Engineering, Low-Latency Systems, OS Internals, Computer Architecture

Networking & Protocols: TCP/IP, UDP, TLS, mTLS, DNS, HTTP/2, HTTP/3, QUIC, WebSockets, Socket Programming, Network Performance Optimization

Network & Cloud Security: eBPF, XDP, Cilium, Kernel-Level Networking, Packet Processing, VPC, Subnets, Routing, NAT, VPN, Peering, Load Balancing, Security Groups / Firewalls

Distributed Systems & Streaming: Distributed Systems (consensus, replication, sharding), Apache Kafka, Apache Flink, Apache Spark, Real-Time Data Pipelines, Change Data Capture (CDC), Microservices

APIs & Serialization: gRPC, REST API Design, Protobuf

ML Inference & AI: PyTorch, TensorFlow, XGBoost, Transformer Models (BERT), LLMs, ONNX Runtime, On-Device/Edge Inference, Model Pruning & Quantization, Computer Vision (YOLOv8/v11), ANN, CNN, NLP, Deep Learning, Recommendation Systems, MLOps

GPU & Parallel Computing: CUDA, CUDA-Oxide (Rust-to-CUDA), PTX, SIMT, GPU Kernel Programming, Parallel Computing, GPU-Accelerated Computing

Cloud & Infra: AWS (EC2, ECS, Lambda, Kinesis, EMR, SageMaker), Azure (AKS, Bicep), Google Cloud, Kubernetes (EKS, AKS), Docker, Terraform, Kubebuilder, Istio, Kubernetes Gateway API

Observability: Prometheus, Grafana, Azure Monitor, AWS CloudWatch

Data & Storage: DynamoDB, PostgreSQL, Redis, MongoDB, MySQL, Amazon Redshift, Amazon S3

Core CS: Data Structures & Algorithms, System Design, Operating Systems, Linux, Computer Architecture

EDUCATION

Bachelor of Technology in Computer Science & Engineering

RCC Institute of Information Technology, Kolkata, India | 2018